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REVIEW

Conversational Agents and Chatbots for Mental Health

A narrative review of the evidence from rule-based systems to large language models

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ABSTRACT

Conversational agents, often called chatbots, are increasingly offered as low-threshold support for people with psychological distress. This narrative review summarizes the development and the empirical evidence of such systems, from early rule-based programs to agents built on large language models. It first describes the technical shift from scripted dialogues to generative systems and the consequences of this shift for predictability and safety. It then discusses early randomized and observational studies of structured agents such as Woebot and Wysa, systematic reviews and meta-analyses of the field, and the first randomized controlled trial of a fully generative therapy chatbot. Overall, the available studies suggest small to moderate short-term reductions in symptoms of depression and distress, together with generally favorable user acceptance. However, the evidence is limited by small samples, short follow-up periods, high attrition, weak comparison conditions such as waitlists or information-only controls, heterogeneous outcome measures, and frequent involvement of developers in the research. Safety in crisis situations and erroneous output from generative models remain insufficiently studied. The review concludes that conversational agents are best understood as a promising but unproven complement to human care, and it identifies priorities for future research.

Keywords: conversational agents · chatbots · digital mental health · large language models · narrative review

1 Introduction

Demand for psychological help exceeds the available supply of trained professionals in many countries, and waiting times for psychotherapy are often long. Digital interventions have been proposed as one way to narrow this gap, and internet-delivered psychological treatments, particularly guided cognitive behavioral programs, now rest on a substantial evidence base (Andersson et al., 2019). Conversational agents represent a more recent development within this field. Instead of presenting modules to be read and completed, they interact with users through dialogue in natural language, usually in text form.

The arrival of systems based on large language models has given the topic new urgency. Such systems can produce fluent and seemingly empathic replies on almost any subject, and many people use general-purpose chatbots for emotional support regardless of whether these were designed for that purpose. The question of what is actually known about the benefits and risks of conversational agents in mental health is therefore of considerable practical relevance.

This narrative review summarizes the main lines of development and evidence. It does not aim at a systematic search or a quantitative synthesis, but draws on key primary studies and published systematic reviews in order to outline the state of knowledge, its limitations, and open questions for research and practice.

2 From rule-based systems to large language models

The history of conversational agents in psychology usually begins with ELIZA, a program that simulated a client-centered therapist by recognizing keywords and transforming the user's statements into questions (Weizenbaum, 1966). Although its mechanism was simple, users readily attributed understanding to it, a phenomenon later referred to as the ELIZA effect.

For a long time, most mental health chatbots followed a rule-based or retrieval-based logic.

Dialogues were scripted by clinicians and developers, users often chose among predefined responses, and the system selected content, for example psychoeducation or exercises derived from cognitive behavioral therapy, according to fixed decision rules. Some systems used natural language processing to classify free-text input, but the content of the replies remained largely predetermined (Laranjo et al., 2018; Vaidyam et al., 2019). This design limits flexibility but makes the system's behavior predictable and auditable.

Generative agents based on large language models work differently. They produce each response anew on the basis of statistical patterns learned from very large amounts of text. This allows open, individualized conversations, but it also means that the content of a given reply cannot be fully specified in advance. Generative models can produce plausible but incorrect or unsupported statements, often described as hallucinations (Ji et al., 2023). For clinical applications, the shift from scripted to generative systems therefore implies a shift in the central questions, from whether a fixed program is effective toward how the behavior of a flexible system can be constrained, monitored, and evaluated (Stade et al., 2024).

3 Early trials and real-world evaluations

One of the most frequently cited early studies is a randomized controlled trial of Woebot, a rule-based agent delivering elements of cognitive behavioral therapy to young adults with symptoms of depression and anxiety (Fitzpatrick et al., 2017). Over a two-week period, participants who used the agent reported a greater reduction in depressive symptoms than participants in an information-only control condition who were directed to an electronic self-help book. The study was small, short, and conducted by researchers connected with the developing company, which the authors acknowledged.

Wysa, an agent combining scripted content with free-text conversation, was evaluated in a real-world mixed-methods study (Inkster et al., 2018). Users with self-reported depressive symptoms who engaged more frequently with the app showed greater average improvement than less engaged users. Because there was no randomized control group and engagement was self-selected, these results cannot be interpreted as evidence of a causal effect. Later work on Woebot reported that users rated their bond with the agent on a working alliance measure at levels comparable to those reported for other forms of therapy, although these findings were also based on observational data from the developer (Darcy et al., 2021).

Safety concerns arose early as well. A study of general-purpose voice assistants found that their responses to statements about suicidal ideation, depression, or interpersonal violence were inconsistent and often incomplete (Miner et al., 2016). Although these assistants were not designed as mental health tools, the findings illustrated how users may turn to conversational systems in crisis situations for which these systems are not prepared.

4 Systematic reviews and meta-analyses

Early systematic reviews described a heterogeneous field with few controlled trials. A review of conversational agents in healthcare concluded that most studies were quasi-experimental or observational and that evidence of clinical effectiveness was limited (Laranjo et al., 2018). A review focused on psychiatry found high user satisfaction and engagement in several studies, but emphasized the preliminary nature of the evidence and the need for more rigorous evaluation (Vaidyam et al., 2019).

A meta-analysis of chatbot interventions for mental health reported some evidence of improvements in depression, distress, and stress, but rated the overall evidence as weak, owing to a small number of studies, high risk of bias, and inconsistent results (Abd-Alrazaq et al., 2020). Few studies reported on safety. A later systematic review and meta-analysis of AI-based conversational agents found reductions in symptoms of depression and psychological distress, whereas effects on psychological well-being were not statistically significant (Li et al., 2023). The authors also explored user experience and suggested that the quality of the human–AI interaction and the content of the dialogue may be relevant for outcomes.

Taken together, these syntheses suggest that conversational agents can produce short-term improvements in symptoms, especially in depression and distress, with effects that appear small to moderate. Broader reviews of digital psychiatry reach a similar assessment and caution that enthusiasm has often outpaced evidence (Torous et al., 2021).

5 The first randomized trial of a generative therapy chatbot

A recent milestone is the first randomized controlled trial of a fully generative AI chatbot designed for mental health treatment (Heinz et al., 2025). The system, called Therabot, was trained on expert-

curated dialogues grounded in evidence-based psychotherapy. Adults with clinically significant symptoms of major depressive disorder, generalized anxiety disorder, or elevated risk for feeding and eating disorders were randomized to several weeks of access to the chatbot or to a waitlist control condition.

The authors reported greater symptom reductions in the intervention group across the three clinical domains, as well as high engagement. Participants' ratings of the therapeutic alliance were described as comparable to those reported for outpatient psychotherapy. Clinicians monitored the conversations, and the team intervened in cases of safety concerns or inappropriate responses.

The trial is important because it shows that a generative system can be evaluated with a randomized design. At the same time, its interpretation is limited. The waitlist comparison does not allow conclusions about efficacy relative to active treatments or to simpler digital programs, the follow-up period was short, and the extensive human monitoring and the specific training procedure mean that the results cannot be transferred to general-purpose chatbots used without supervision.

6 Methodological limitations and open questions

Across the literature, several limitations recur. Samples are often small and drawn from convenience populations such as students or app users. Follow-up periods are short, so that the durability of effects is largely unknown. Attrition is frequently high, and the handling of missing data varies. Comparison conditions are mostly waitlists or information-only controls, which tend to overestimate effects relative to active alternatives. Outcome measures are heterogeneous, and many studies are conducted or funded by the developers of the respective product (Abd-Alrazaq et al., 2020; Li et al., 2023).

For generative systems, additional problems arise. Model versions change quickly, which makes replication difficult and means that a published trial may no longer describe the system that users actually encounter. Erroneous or harmful output cannot be excluded by design (Ji et al., 2023), and crisis detection and referral pathways are rarely evaluated systematically. Proposals for responsible development therefore recommend staged evaluation, from safety testing to controlled trials, with clinical expertise involved throughout (Stade et al., 2024).

Finally, little is known about the mechanisms through which conversational agents might work. Whether perceived bonds with an agent contribute to outcomes in the same way as the alliance in human psychotherapy, and what role meaning, responsibility, and the user's own reflection play, are questions that require both empirical and conceptual work.

7 Conclusion

The evidence on conversational agents for mental health has grown considerably over the past decade. Studies of rule-based agents and meta-analyses suggest modest short-term benefits for symptoms of depression and distress, and the first trial of a generative therapy chatbot has yielded encouraging results under carefully controlled conditions. Yet the evidence remains limited by short follow-up, weak comparators, high attrition, and potential conflicts of interest, and safety questions are far from resolved.

Conversational agents are therefore best regarded as a promising but unproven complement to human care rather than as a substitute for it. Future research should use active comparison conditions, longer follow-up, transparent reporting of model versions and safety events, and independent evaluation. For a meaning-centered psychology, it will also be important to ask not only whether symptoms improve, but whether such tools strengthen or weaken a person's capacity to take a stance toward their own situation.

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